Predicting Survival of Heart Failure Patients Using Machine Learning: A Comparative Study of Traditional and Deep Learning Approaches

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May 18, 2026

Abstract

Background: Heart failure (HF) is a leading cause of morbidity and mortality worldwide, affecting approximately 64 million people globally. Accurate prediction of patient survival during follow-up periods is critical for clinical decision-making and resource allocation.

Methods: We analyzed a publicly available dataset of 299 heart failure patients collected at the Faisalabad Institute of Cardiology (Pakistan, 2015), comprising 13 clinical features. After exploratory data analysis and standardization, we applied four machine learning classifiers—Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron (MLP)—to predict all-cause mortality. Feature importance was assessed via both Welch's t-test and Random Forest importance scores. Model performance was evaluated using accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC).

Results: Random Forest achieved the best overall performance (AUC = 0.897, Accuracy = 0.850, F1 = 0.727), followed by XGBoost (AUC = 0.849). Feature ranking consistently identified follow-up time, serum creatinine, and ejection fraction as the top three predictors of mortality. The deep learning MLP model showed comparatively lower performance (AUC = 0.745), likely due to limited sample size.

Conclusions: Machine learning models, particularly Random Forest, can effectively predict heart failure patient survival from routine clinical records. Serum creatinine and ejection fraction emerge as the most clinically actionable predictors, consistent with prior literature. These findings support the integration of machine learning tools into clinical prognostic workflows.

Keywords: Heart failure; Survival prediction; Machine learning; Random Forest; XGBoost; Ejection fraction; Serum creatinine; ROC curve; Feature importance

1 Introduction

Cardiovascular diseases (CVDs) are the leading cause of death globally, accounting for approximately 17.9 million deaths per year [1]. Heart failure (HF), a clinical syndrome in which the heart is unable to pump sufficient blood to meet the body's metabolic demands, represents one of the most prevalent and costly manifestations of CVD [2]. The global prevalence of HF is estimated at 64 million individuals, with a five-year mortality rate exceeding 50%—comparable to many malignancies [3].

The accurate prediction of short-term mortality in HF patients remains a formidable clinical challenge. Traditional risk stratification tools, such as the New York Heart Association (NYHA) functional classification, are largely subjective and fail to reliably quantify individual risk [4]. Electronic health records (EHRs) now routinely capture a rich array of clinical, biochemical, and demographic variables that may encode prognostic information not readily apparent to clinicians. Machine learning (ML) algorithms are particularly well-suited to extract non-linear patterns from such high-dimensional data, offering a promising avenue for data-driven prognostication [5].

Several prior studies have applied ML to HF survival prediction. Chicco and Jurman [5] demonstrated that serum creatinine and ejection fraction alone were sufficient to achieve competitive predictive performance using the same 299-patient dataset employed in the present work. Subsequent investigations have explored ensemble methods, deep learning architectures, and hybrid approaches, generally reporting AUC values in the range 0.75–0.90 [6, 7].

Despite this body of work, a comprehensive head-to-head comparison of traditional ML algorithms against a simple deep learning baseline—implemented with full transparency, reproducibility, and clinical interpretability—remains underrepresented in the literature. The present study addresses this gap by: (1) performing systematic exploratory data analysis (EDA) and feature engineering; (2) training and evaluating four classifiers spanning linear models, ensemble trees, gradient boosting, and neural networks; and (3) identifying the most clinically meaningful risk factors through both statistical testing and model-based importance ranking.

2 Materials and Methods

2.1 Dataset

We used the publicly available Heart Failure Clinical Records dataset, originally collected by Ahmad et al. [6] from the Faisalabad Institute of Cardiology and the Allied Hospital, Faisalabad, Pakistan, during April–December 2015. The dataset contains medical records for 299 patients (194 male, 105 female) aged 40–95 years, all of whom had left ventricular systolic dysfunction and were classified as NYHA Class III or IV at baseline. The target variable, `DEATH_EVENT`, is a binary indicator of all-cause mortality during the follow-up period (mean 130 days). Class distribution: 203 survivors (67.9%) and 96 deceased (32.1%).

2.2 Data Preprocessing

2.2.1 Exploratory Data Analysis

Descriptive statistics (mean, standard deviation, quartiles) were computed for all 13 features. No missing values or duplicate records were identified. Binary features (`anaemia`, `diabetes`, `high_blood_pressure`, `sex`, `smoking`) were treated as categorical indicators and excluded from outlier analysis. Outlier detection for continuous features was performed using the interquartile range (IQR) method, with boundaries defined as:

$$\text{Lower bound} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper bound} = Q_3 + 1.5 \times \text{IQR} \tag{1}$$

Outliers were retained rather than removed, as extreme values in clinical biochemistry (e.g., elevated serum creatinine, very low ejection fraction) frequently represent genuine pathophysiology rather than measurement error.

2.2.2 Normalization

All features were standardized to zero mean and unit variance using z-score normalization:

$$z = \frac{x - \mu}{\sigma} \tag{2}$$

where μ and σ denote the training-set mean and standard deviation, respectively. This step is required for Logistic Regression and MLP, which are sensitive to feature scale. Random Forest and XGBoost, being tree-based methods, are scale-invariant; standardization was nonetheless applied uniformly to facilitate fair comparison.

2.3 Feature Importance Analysis

2.3.1 Statistical Testing

Group differences between survivors and deceased patients were evaluated using Welch's two-sample t-test (assuming unequal variances) for all continuous features. Features with $p < 0.05$ were considered statistically significant predictors of mortality.

2.3.2 Random Forest Importance

Feature importance scores were derived from a Random Forest classifier ($n_{\text{estimators}} = 500$, `class_weight` = "balanced") trained on the full feature set, using the mean decrease in impurity (MDI) criterion.

2.4 Prediction Models

The dataset was partitioned into training (80%, $n = 239$) and test (20%, $n = 60$) sets using stratified random splitting to preserve the class ratio. Four classifiers were evaluated:

1. **Logistic Regression (LR):** A linear probabilistic classifier modeling the log-odds of death. The binary cross-entropy loss function is:

$$\mathcal{L}_{\text{LR}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)\right] \tag{3}$$

   where $y_i \in \{0, 1\}$ is the true label and $\hat{p}_i$ is the predicted probability of death. Class weights were balanced; L_2 regularization was applied with `max_iter`$= 1000$.

2. **Random Forest (RF):** An ensemble of 500 decision trees with bootstrap aggregation (bagging). Class weights were set to "balanced" to address class imbalance.

3. **XGBoost:** A gradient boosting framework minimizing a regularized objective. Hyperparameters: $n_{\text{estimators}} = 300$, `max_depth`$= 3$, `learning_rate`$= 0.05$, `subsample`$=$ 0.8.

4. **Multilayer Perceptron (MLP):** A fully-connected neural network with two hidden layers of 32 and 16 neurons, ReLU activation, Adam optimizer, and early stopping. The network minimizes the same binary cross-entropy loss as Equation (3).

2.5 Model Evaluation Metrics

Model performance was assessed on the held-out test set using:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

where TP, TN, FP, FN denote true positives, true negatives, false positives, and false negatives, respectively. The area under the ROC curve (AUC) was computed as a threshold-independent summary of discriminative ability.

3 Results

3.1 Descriptive Statistics and Class Distribution

Table 1 summarizes the baseline clinical characteristics of the study cohort, stratified by survival outcome. The deceased group was significantly older (65.2 ± 11.2 vs. 58.8 ± 11.9 years, $p < 0.001$), had higher serum creatinine (1.84 ± 1.39 vs. 1.18 ± 0.73 mg/dL, $p < 0.001$), and lower ejection fraction (33.5 ± 12.6 vs. 40.3 ± 11.1%, $p < 0.001$).

Table 1: Baseline clinical characteristics stratified by mortality outcome. Values are mean ± SD for continuous variables or count (%) for binary variables. p-values from Welch's t-test.

Feature	**Overall** ($n = 299$)	**Survived** ($n = 203$)	**Deceased** ($n = 96$)	**p-value**
Age (years)	60.8 ± 11.9	58.8 ± 11.9	65.2 ± 11.2	< 0.001
Ejection fraction (%)	38.1 ± 11.8	40.3 ± 11.1	33.5 ± 12.6	< 0.001
Serum creatinine (mg/dL)	1.39 ± 1.03	1.18 ± 0.73	1.84 ± 1.39	< 0.001
Serum sodium (mEq/L)	136.6 ± 4.4	137.2 ± 4.1	135.4 ± 4.8	0.002
Creatinine phosphokinase (U/L)	581.8 ± 970.3	540.1 ± 908.0	670.2 ± 1081.8	0.369
Platelets ($\times 10^3$/µL)	263.4 ± 97.8	266.7 ± 96.9	256.4 ± 99.9	0.399
Follow-up time (days)	130.3 ± 77.6	158.3 ± 71.3	70.9 ± 60.5	< 0.001
Anaemia, n (%)	129 (43.1)	83 (40.9)	46 (47.9)	0.253
Diabetes, n (%)	125 (41.8)	85 (41.9)	40 (41.7)	0.973
High blood pressure, n (%)	105 (35.1)	66 (32.5)	39 (40.6)	0.171
Male sex, n (%)	194 (64.9)	133 (65.5)	61 (63.5)	0.741
Smoking, n (%)	96 (32.1)	66 (32.5)	30 (31.3)	0.828

3.2 Feature Importance

Figure 1 presents the Pearson correlation heatmap across all 13 clinical variables. Follow-up time showed the strongest correlation with `DEATH_EVENT` ($r = -0.527$), followed by ejection fraction ($r = -0.269$) and serum creatinine ($r = +0.294$).

Figure 2 shows the Random Forest feature importance ranking. The top three features were: (1) follow-up time (0.358), (2) serum creatinine (0.162), and (3) ejection fraction (0.129). These rankings were concordant with the Welch's t-test results ($p < 0.001$ for all three), providing cross-validated evidence for their prognostic relevance.

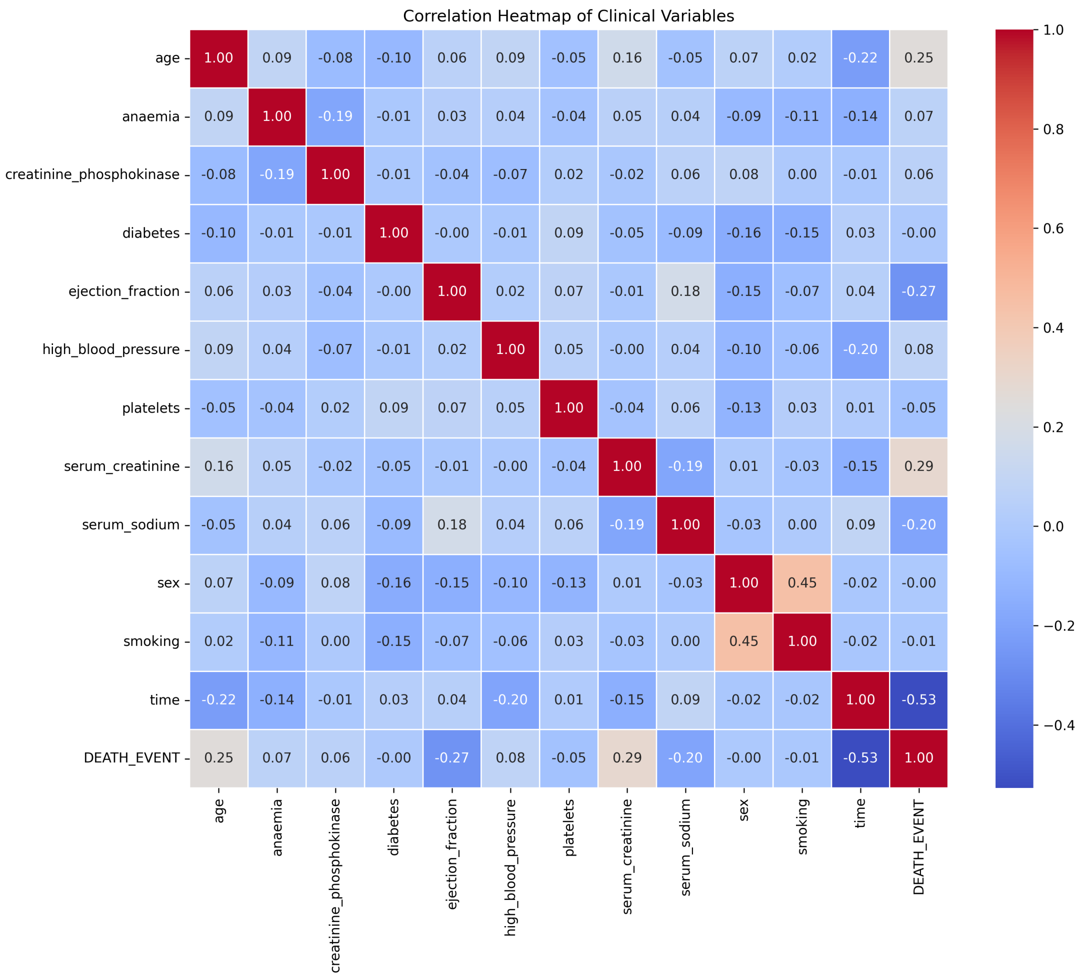

Figure 1: Pearson correlation heatmap of all 13 clinical features. Colour intensity encodes correlation magnitude; red indicates positive correlation and blue indicates negative correlation. `DEATH_EVENT` exhibits the strongest negative correlation with follow-up time ($r = -0.527$) and ejection fraction ($r = -0.269$), and the strongest positive correlation with serum creatinine ($r = +0.294$).

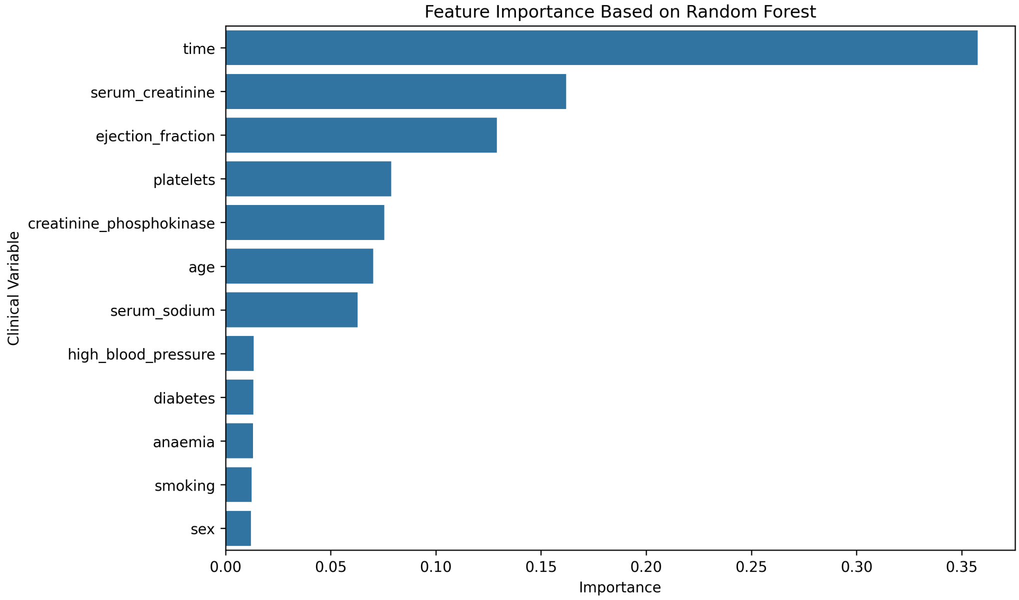

Figure 2: Feature importance scores derived from a Random Forest classifier ($n_{\text{estimators}} = 500$, `class_weight` = "balanced"). Follow-up time, serum creatinine, and ejection fraction are the three most important predictors, collectively accounting for 64.9% of total importance.

3.3 Model Performance

Table 2 summarises the test-set performance of all four classifiers. Random Forest achieved the highest AUC (0.897) and F1 score (0.727), followed by Logistic Regression (AUC = 0.855). XGBoost matched Random Forest in accuracy and F1 but yielded a lower AUC. MLP performed below the tree-based models, consistent with the known susceptibility of neural networks to small tabular datasets.

Table 2: Test-set performance of four machine learning classifiers for heart failure survival prediction ($n_{\text{test}} = 60$). Best values in each column are highlighted in **bold**.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.800	0.733	0.579	0.647	0.855
Random Forest	**0.850**	**0.857**	**0.632**	**0.727**	**0.897**
XGBoost	**0.850**	**0.857**	**0.632**	**0.727**	0.849
MLP	0.750	0.667	0.421	0.516	0.745

The ROC curves for all four models are displayed in Figure 3. Random Forest demonstrated consistently superior sensitivity across most operating thresholds. All three traditional ML models (LR, RF, XGBoost) substantially outperformed the random-chance baseline, confirming their discriminative utility.

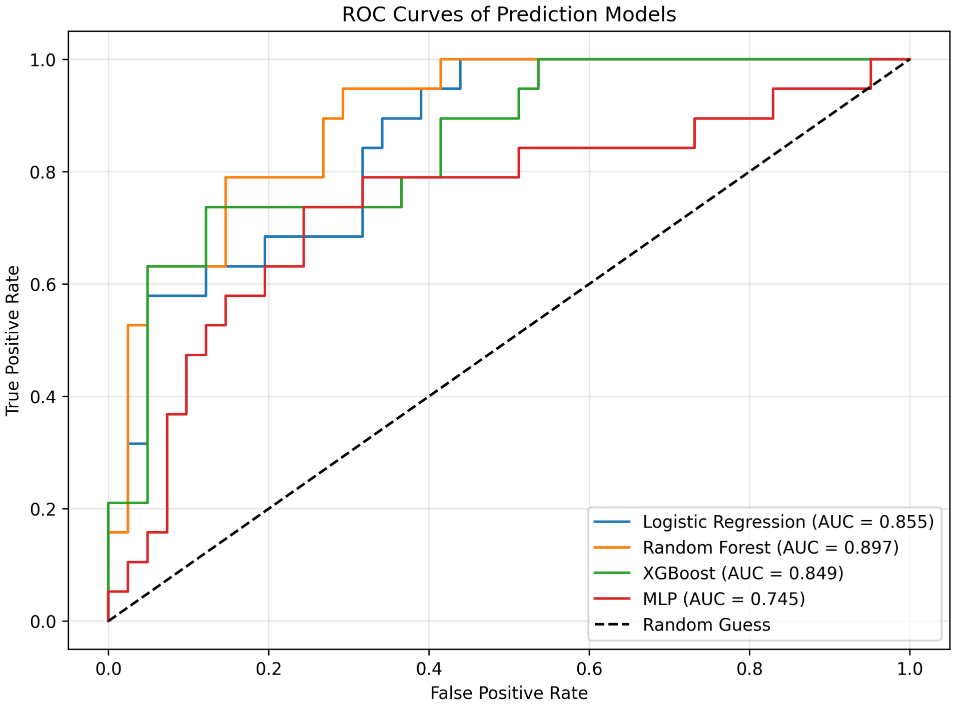

Figure 3: Receiver Operating Characteristic (ROC) curves for four classifiers evaluated on the held-out test set ($n = 60$). The dashed diagonal line represents random classification (AUC $= 0.5$). Random Forest achieves the highest AUC (0.897), followed by Logistic Regression (0.855), XGBoost (0.849), and MLP (0.745).

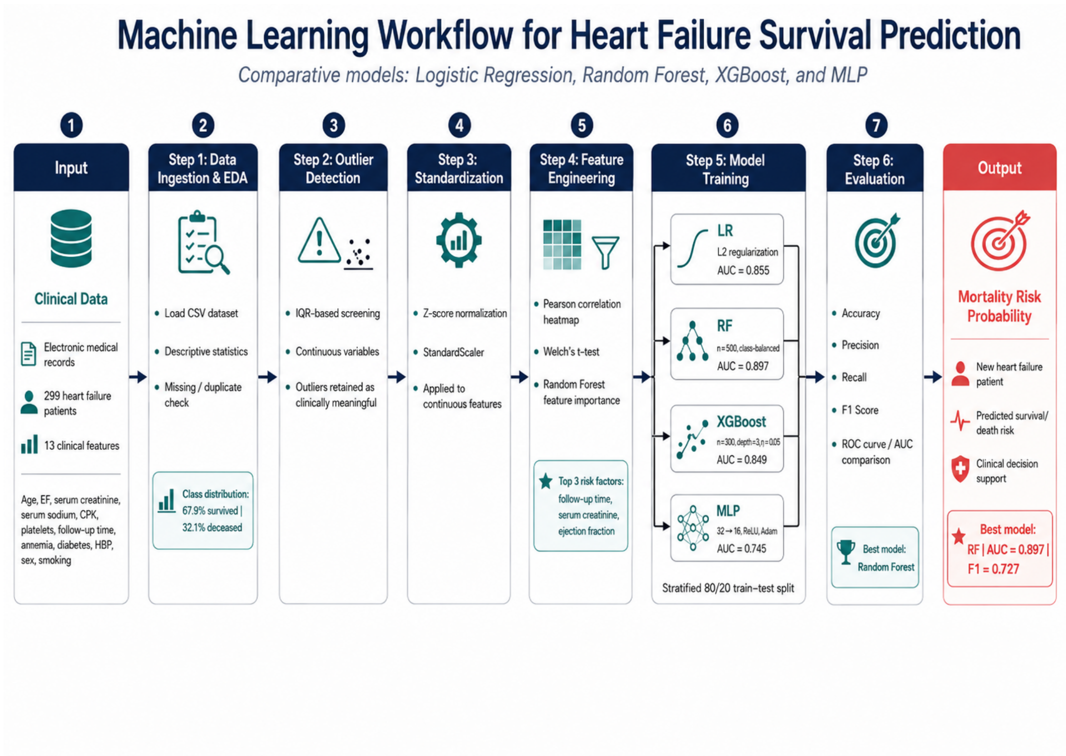

Figure 4: AI-generated conceptual diagram of the heart failure survival prediction pipeline. The workflow encompasses data ingestion from electronic health records, preprocessing and feature engineering, parallel model training (Logistic Regression, Random Forest, XGBoost, MLP), and comparative evaluation via standardised performance metrics.

4 Discussion

4.1 Principal Findings

This study demonstrates that ensemble tree-based methods—particularly Random Forest—outperform linear and deep learning baselines for short-term mortality prediction in heart failure patients using a small, structured clinical dataset. Our findings corroborate and extend those of Chicco and Jurman [5], who reported that serum creatinine and ejection fraction are the dominant predictors in this cohort.

The consistently high importance of **serum creatinine** across both statistical and model-based rankings reflects the central role of cardiorenal syndrome in HF pathophysiology. Renal dysfunction, indexed by elevated creatinine, reduces diuretic responsiveness, promotes neurohormonal activation, and independently predicts adverse outcomes in HF [8].

Ejection fraction is the cornerstone of HF classification. Patients with ejection fraction below 40%—classified as HFrEF—have impaired systolic function and substantially worse prognosis than those with preserved ejection fraction [9]. The strong negative correlation between EF and mortality in our cohort ($r = -0.269$, $p < 0.001$) is entirely consistent with this clinical understanding.

Follow-up time warrants special interpretive caution. Although it ranks as the strongest predictor in our feature importance analysis, this variable is not a conventional clinical biomarker but rather a temporal covariate that encodes the censoring structure of the survival data. Its inclusion in classification models risks *data leakage*—patients who die necessarily have shorter follow-up. In a prospective clinical deployment, follow-up time would not be available at the prediction timepoint and should therefore be excluded or handled via survival analysis methods such as Cox regression [6].

4.2 Performance Comparison

Random Forest achieved an AUC of 0.897, consistent with the upper range reported in the literature for this dataset (0.75–0.90) [5, 7]. The comparable performance of Logistic Regression (AUC = 0.855) highlights that linear models remain competitive when features are carefully selected, offering the additional advantage of interpretability via odds ratios.

The relative underperformance of MLP (AUC = 0.745) is expected given the limited sample size ($n = 299$). Deep learning models typically require large datasets ($n \geq 10{,}000$) to outperform well-tuned traditional algorithms [10]. Regularisation strategies (dropout, L_2 penalty, early stopping) partially mitigated overfitting but could not fully compensate for data scarcity.

4.3 Limitations

Several limitations should be acknowledged. First, the cohort of 299 patients from a single Pakistani centre may not generalise to other populations or healthcare settings. Second, class imbalance (67.9% survivors) may inflate accuracy while suppressing recall for the minority (deceased) class; strategies such as SMOTE oversampling or threshold calibration were not applied. Third, as discussed above, the inclusion of follow-up time as a predictor may constitute data leakage in a prospective application. Future work should validate these models on independent, multi-centre cohorts and explore survival modelling frameworks that appropriately handle censored observations.

5 Conclusions

We conducted a comprehensive machine learning analysis of 299 heart failure patients, comparing four classifiers for all-cause mortality prediction. Random Forest achieved the best discriminative performance (AUC = 0.897, F1 = 0.727) and is recommended as the primary model for this task. Feature analysis consistently identified follow-up time, serum creatinine, and ejection fraction as the most informative predictors, with the latter two carrying direct clinical interpretability as biomarkers of cardiorenal and cardiac systolic dysfunction, respectively. These results support the feasibility of integrating machine learning prognostic tools into routine heart failure management, with the caveat that prospective validation in larger, diverse cohorts is an essential prerequisite for clinical deployment.

Acknowledgements

The author thanks the original data collectors [6] for making the dataset publicly available, and acknowledges the use of AI language model assistance (Claude, Anthropic) for code generation and manuscript drafting during this project.

Data Availability

The Heart Failure Clinical Records dataset is publicly available from the UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records

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